

**MOHAMMAD NASSER IBRAHIM** UAE National (Family Book: Dubai) | 050-389-8770 | mohammadni.cse@gmail.com | [github.com/mohammadnicse](https://github.com/mohammadnicse) | Dubai, UAE

**PROFESSIONAL SUMMARY** Computer Science student at the American University of Sharjah (AUS) and UAE National (Dubai). ATHS graduate with a strong technical discipline in software logic and systems engineering. Experienced in applying C++ and Python to systems engineering, data exploration, and software logic. Eager to transition these academic foundations into the fields of serious games and simulation environments. Seeking to contribute to the Dubai Police Virtual Technology Center by leveraging data analysis and object-oriented programming to build and evaluate interactive training scenarios.

**EDUCATION American University of Sharjah (AUS)** | Sharjah, UAE *Bachelor of Science in Computer Science* | Expected Graduation: Spring 2027

**Applied Technology High School (ATHS)** | Dubai, UAE *High School Diploma (Advanced Science/Engineering)* | Grade: 90.6%

## TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL, Bash Scripting, Assembly.
- **Simulation & Game Logic:** Object-Oriented Programming (OOP), State Machines, 2D Collision Logic, Branching Scenarios.
- **Data & Visualization:** Streamlit (Dashboards), Scikit-Learn (Basic Data Modeling), Matplotlib.
- **Systems & Tools:** Linux (Ubuntu), Git/GitHub, Memory Management, Hardware-Software Integration.

## ACADEMIC & PERSONAL PROJECTS

### Interactive Scenario & Decision Simulator | *Python, Streamlit*

- Developed a branching-logic simulation prototype designed to track and evaluate user decision-making in a digital environment.
- Built an interactive dashboard to capture real-time user inputs, logging success rates and decision pathways across different scenario branches.

### C++ Physics-Logic Engine Prototype (2D) | *C++, Linux*

- Programmed a modular, lightweight 2D engine to simulate object interactions, basic collisions, and real-time state changes.
- Optimized memory management during the simulation loop using custom allocation techniques to prevent system lag during entity generation.

### Training Data Exploration & Visualization (ICS-Sentinel) | *Python, Scikit-Learn*

- Developed the Python backend and Streamlit dashboard for a group project focused on parsing and visualizing complex datasets.
- Utilized basic Machine Learning models on the SWaT dataset to visually differentiate between standard operational data and simulated anomalies.

### System-Level Memory Guard (HeapGuard) | *C++, Linux*

- Developed a custom C++ memory allocator to manage data safety and resource zeroing at the system level.
- Implemented guard bands to detect data corruption during runtime, demonstrating a deep understanding of how software interacts with system hardware.

## VOLUNTEERING Fazaa Volunteering | Active Volunteer (2018 - 2020)

- Participated in various community initiatives across the UAE.
- Developed strong teamwork, communication, and organizational skills while working in structured public-facing environments.